

Differentiating Implicit Relations

Answer Key

AP Calculus AB/BC

Quick Answers

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|--|---|
| 1. (a) the derivative of the upper branch = $\frac{-x}{\sqrt{25-x^2}}$, (b) its value at $x = 3 = -\frac{3}{4}$ | 5. (a) the derivative of the upper branch = $\frac{1}{\sqrt{x}}$,
(b) the implicit slope at the point = 1 |
| 2. $-\frac{3}{4}$ | 6. -1 |
| 3. $-\frac{4}{5}$ | 7. $-\frac{4}{3}$ |
| 4. $-\frac{1}{4}$ | 8. (a) the slope at the point = -1 , (b) the symmetry argument = — |
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What is verified

10 of 11 answers machine-checked against SymPy, across 8 problems.
 — marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-3.D – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 11 tagged checks.

Common wrong answers

6. If they got 0: differentiated the first term as x squared times y prime only, dropping the $2xy$ piece the product rule contributes

1. Circle $x^2 + y^2 = 25$ near $(3, 4)$, upper branch $y = \sqrt{25 - x^2}$. (a) $\frac{dy}{dx}$ in terms of x . (b) Its value at $x = 3$.

Solution (a): Write the branch as $(25 - x^2)^{1/2}$ and use the chain rule:

$$\frac{dy}{dx} = \frac{1}{2} (25 - x^2)^{-1/2} (-2x) = \frac{-x}{\sqrt{25 - x^2}}.$$

So the derivative is

Solution (b): At $x = 3$ the radical is $\sqrt{25 - 9} = 4$, so the slope is $-\frac{3}{4}$.

$\frac{-x}{\sqrt{25 - x^2}}$
$-\frac{3}{4}$

That 4 in the denominator is exactly the y -coordinate of the point — which is the clue that the implicit form will be simpler.

2. Same circle, implicitly: differentiate $x^2 + y^2 = 25$ and evaluate at $(3, 4)$.

Solution: Differentiate both sides with respect to x , remembering the chain-rule factor on y^2 :

$$2x + 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}.$$

At $(3, 4)$ this is $-\frac{3}{4}$.

$-\frac{3}{4}$

Same tangent line as problem 1, reached in two lines instead of a chain-rule computation, and valid on the lower half of the circle too. Substituting the coordinates the wrong way round gives $-\frac{4}{3}$, which is the reciprocal and is the error to watch for.

3. $x^2 + xy + y^2 = 7$ at $(1, 2)$.

Solution: Check the point first: $1 + 2 + 4 = 7$. Now differentiate, using the product rule on xy :

$$2x + \left(y + x \frac{dy}{dx}\right) + 2y \frac{dy}{dx} = 0$$

$$(x + 2y) \frac{dy}{dx} = -(2x + y)$$

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}.$$

At $(1, 2)$: $-\frac{2+2}{1+4} = -\frac{4}{5}$.

$-\frac{4}{5}$

4. $x^3 + y^3 = 9$ at $(1, 2)$.

Solution: The point checks: $1 + 8 = 9$. Differentiating,

$$3x^2 + 3y^2 \frac{dy}{dx} = 0 \quad \implies \quad \frac{dy}{dx} = -\frac{x^2}{y^2}.$$

At $(1, 2)$: $-\frac{1}{4}$.

$-\frac{1}{4}$

Dropping the squares and writing $-x/y$ would give $-\frac{1}{2}$; the squares come from differentiating cubes and cannot be dropped.

5. Parabola $y^2 = 4x$ at $(1, 2)$. (a) Derivative of the branch $y = 2\sqrt{x}$. (b) The implicit slope at $(1, 2)$.

Solution (a): $y = 2x^{1/2}$, so $\frac{dy}{dx} = 2 \cdot \frac{1}{2}x^{-1/2} = \frac{1}{\sqrt{x}}$.

$$\frac{1}{\sqrt{x}}$$

Solution (b): Implicitly, $2y\frac{dy}{dx} = 4$, so $\frac{dy}{dx} = \frac{2}{y}$. At $(1, 2)$ that is $\frac{2}{2}$.

$$1$$

Part (a) at $x = 1$ gives $\frac{1}{\sqrt{1}} = 1$ as well: the two forms $\frac{1}{\sqrt{x}}$ and $\frac{2}{y}$ are the same function on this branch, because $y = 2\sqrt{x}$ there.

6. $x^2y + y^2 = 12$ at $(2, 2)$.

Solution: The point checks: $4(2) + 4 = 12$. The first term is a product, so

$$\begin{aligned} 2xy + x^2\frac{dy}{dx} + 2y\frac{dy}{dx} &= 0 \\ (x^2 + 2y)\frac{dy}{dx} &= -2xy \\ \frac{dy}{dx} &= -\frac{2xy}{x^2 + 2y}. \end{aligned}$$

At $(2, 2)$: $-\frac{8}{4+4} = -\frac{8}{8}$.

$$-1$$

Skipping the product rule — differentiating x^2y as $x^2\frac{dy}{dx}$ alone — makes the numerator vanish and produces a slope of 0. A horizontal tangent here would be visible nonsense, which is a useful check.

7. $x^2y^3 = 8$ at $(1, 2)$.

Solution: The point checks: $1 \cdot 8 = 8$. Product rule on the left, with the chain rule inside y^3 :

$$\begin{aligned} 2xy^3 + x^2\left(3y^2\frac{dy}{dx}\right) &= 0 \\ \frac{dy}{dx} &= -\frac{2xy^3}{3x^2y^2} = -\frac{2y}{3x}. \end{aligned}$$

At $(1, 2)$: $-\frac{4}{3}$.

$$-\frac{4}{3}$$

The simplification from $\frac{2xy^3}{3x^2y^2}$ to $\frac{2y}{3x}$ is worth insisting on — it makes the substitution a one-line job.

8. Challenge. Folium $x^3 + y^3 = 6xy$ at $(3, 3)$. (a) The slope. (b) The symmetry argument.

Solution (a): The point checks: $27 + 27 = 54$ and $6(9) = 54$. Differentiate both sides, product rule on the right:

$$\begin{aligned} 3x^2 + 3y^2\frac{dy}{dx} &= 6y + 6x\frac{dy}{dx} \\ (3y^2 - 6x)\frac{dy}{dx} &= 6y - 3x^2 \\ \frac{dy}{dx} &= \frac{2y - x^2}{y^2 - 2x}. \end{aligned}$$

At $(3, 3)$: $\frac{6-9}{9-6} = \frac{-3}{3}$.

-1

Solution (b) — instructor-judged, no machine check. Swapping x and y leaves $x^3 + y^3 = 6xy$ unchanged, so the curve is symmetric about the line $y = x$. The point $(3, 3)$ lies on that mirror line, so reflecting the curve carries the tangent at $(3, 3)$ to itself. A reflection in $y = x$ turns a slope m into $\frac{1}{m}$, so $m = \frac{1}{m}$ and $m = \pm 1$. It cannot be $+1$, because that is the direction of the mirror line itself and the curve crosses $y = x$ at $(3, 3)$ rather than running along it — leaving $m = -1$, which is what part (a) computed. Full credit needs the symmetry observation and the conclusion that the slope must be its own reciprocal; award partial credit for noticing the symmetry alone.